



Anil Neerukonda Institute of Technology & Sciences (Autonomous)

Affiliated to AU, Approved by AICTE & Accredited by NAAC with A+ Grade

Accredited by NBA (B.Tech – ECE, EEE, CSE, IT, MECH, Civil & Chemical)

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Department of CSE(AI&ML)

Course Code	Title of the course	Category	L	T	P	Total Instruction Hours	TW + SL	Total Hours	Sessional Marks	End Exam Marks	Total Marks	Credits
23CM9205	DevOps	PE	1	0	3	30	45	75	60	40	100	2.5

Prerequisites:

- Operating System,
- Virtualization,
- Cloud Computing
- Java and Web Programming
- Software Engineering
- Monitor the Software Applications

Course Objectives:

- DevOps improves collaboration and productivity by automating infrastructure and workflows and continuously measuring applications performance.

Course Outcomes:

At the end of the course the student will be able to:

CO	Course outcome Statement
CO-1	Explain DevOps principles and apply related technologies for coding, building, testing, configuring, and monitoring software applications.
CO-2	Implement Continuous Integration/Continuous Deployment (CI/CD) pipelines using Jenkins to automate testing and deployment of Java/Web applications. (CI/CD Automation)
CO-3	Manage version control systems (Git/Mercurial) and containerized environments (Docker) for efficient software development and deployment. (Version Control & Containerization)
CO-4	Automate infrastructure provisioning and configuration management using tools like Ansible/Puppet/Chef for scalable and reproducible deployments. (Infrastructure as Code)

CO-PO Mapping:

COs	Program Outcomes (POs)												
	Domain Specific POs					Domain Independent POs						PSO	
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2	-	2	2	2	-	-	-	2	-	-	1	1
CO2	2	2	3	2	3	-	-	-	2	1	-	1	1
CO3	2	2	3	2	3	-	-	-	2	1	-	1	1
CO4	2	2	3	3	3	-	-	-	2	1	2	2	2

Correlation levels1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

SYLLABUS

PC with following Configuration:

1. Windows or Linux Desktop OS for Client machines
2. Internet Connection for each PC with at least 2 MBPS
3. Intel Core i3/i5/i7 Processor with Intel VT-X support
4. GB RAM Minimum
5. 500 GB Hard Disk
6. Gigabit Ethernet (GbE) network interface card
(NIC CentOS/Fedora/Ubuntu/Red hat Server OS
for One Server)
7. JDK or higher
8. NetBeans or Eclipse
9. OpenSSH.
10. List of Software 's - Maven, Jenkins

Week -1 & 2:

Introduction to DevOps: Emergence of DevOps, Core Principles of DevOps, Essence of a DevOps Culture, Key Elements of DevOps.

1: To Understand the Concept of DevOps with related technologies which are used to Code, Build, Test, Configure & Monitor the Software Applications (CO1)

Week- 3 & 4:

Introduction to Jenkins: Concept of Jenkins and working process.

2: To Install and Configure Jenkins to test, and deploy Java or Web Applications using NetBeans or eclipse. (CO2)

Week 5&6:

Introduction to GIT and its commands, CVS.

3: To Perform Version Control on websites/ Software's using different Version control tools like GIT/Mercurial (Any one) (CO3)

Week 7:

Introduction to Docker and containers, how to upload images to Docker and create containers.

4: To Install and Configure Docker for creating Containers of different Operating System Images (CO3)

Week 8:

5: To build, deploy and manage web or Java application on Docker (CO3)

Week 9:

What is Chef, applications and creation of chef.

6: To install and configure Software Configuration Management using Chef (CO4)

Week 10:

What is Puppet, application, Limitations of Puppet.

7: To install and configure Software Configuration Management using Puppet (CO4)

Week 11:

Concept of Ansible, limitations, its applications.

8: To install and configure Software Configuration Management using Ansible (CO4)

Week 12:

Concept of Salt stack and working process, limitations and its applications.

9: To install and configure Software Configuration Management using Salt stack. (CO4)

Text Book:

1. ProGit, Scott Chacon and Ben Straub, Apress, 2nd Edition 2014. (Exp: 1, 3)
2. Docker: Up and Running Karl Matthias & Sean P. Kane, O'Reilly Publication. (Exp: 1, 4, 5)
3. DevOps, A Software Architects Perspective Len Dass, IngoWeber, LimingZhu, Addison Wesley-Pearson Publication. (Exp: 1, 6, 7)
4. The Definitive Guide to Jenkins John Ferguson Smart, Jenkins, O'Reilly Publication. (Exp: 1, 2)

Reference Book:

1. DevOps for Dummies Sanjeev Sharma and Bernie Coyne, Wiley Publication
2. DevOps for Developers, Huttermann, Michael, Apress Publication.
3. Practical DevOps, Joakim Verona, Pack publication.

PSO: CSE(AI&ML)

PROGRAM SPECIFIC OUTCOMES (PSOs):

PSO 1: Apply Artificial Intelligence and Machine Learning concepts for the development of intelligent systems and applications to multidisciplinary problems.

PSO 2: Develop and implement automated solutions that leverage computing techniques to address real-world and societal problems.

Mapping of Course Outcomes with Program Outcomes – DevOps(23CM9205):

Course outcome	PO Mapped	Level Mapped	Justification for Mapping
CO-1	PO1	2	- The student is able to apply the knowledge of operating systems, virtualization, and cloud computing in understanding DevOps workflows (Competencies:1.2.1). - The student is able to apply computer science principles to understand DevOps lifecycle and automation processes (Competencies:1.7.1).
	PO3	2	- The student is able to define problem statements related to software development and deployment workflows (Competencies:3.5.1). - The student is able to identify system requirements for DevOps environments and tools (Competencies:3.5.2).
	PO4	2	- The student is able to analyze DevOps processes and identify bottlenecks in development and deployment pipelines (Competencies:4.4.1). - The student is able to select appropriate tools and procedures for improving DevOps efficiency (Competencies:4.4.2).
	PO5	2	The student is able to identify modern tools such as Jenkins, Git, and Docker for DevOps practices (Competencies:5.4.1).
	PO9	2	- The student is able to interpret technical documentation related to DevOps tools and workflows (Competencies:9.4.1). - The student is able to document DevOps processes clearly (Competencies:9.4.2).
	PSO1	1	The student is able to analyze real-world problems and identify DevOps-based automation solutions.
	PSO2	1	The student is able to identify societal and industrial problems that can be solved using automated DevOps pipelines.
	PO1	2	- The student is able to apply software engineering principles to automate testing and deployment processes (Competencies:1.2.1). - The student is able to apply computing fundamentals to CI/CD pipeline implementation (Competencies:1.7.1).
	PO2	2	The student is able to analyze CI/CD problems and identify

CO-2			automation strategies (Competencies:2.5.1). The student is able to break down deployment systems into interconnected modules (Competencies:2.6.1).
	PO3	3	- The student is able to design CI/CD pipelines to meet functional requirements of applications (Competencies:3.6.2). - The student is able to evaluate and refine pipeline architectures (Competencies:3.7.1). - The student is able to implement automated deployment workflows using Jenkins (Competencies:3.8.1).
	PO4	2	The student is able to analyze CI/CD performance and improve system efficiency (Competencies:4.6.1).
	PO5	3	- The student is able to demonstrate proficiency in using Jenkins and automation tools (Competencies:5.5.2). - The student is able to evaluate tool limitations in CI/CD environments (Competencies:5.5.1).
	PO9	2	The student is able to present CI/CD workflows and results effectively (Competencies:9.5.2).
	PO10	1	The student is able to manage tasks and scheduling in CI/CD pipelines (Competencies:10.3.2).
	PSO1	1	The student is able to develop automated solutions using DevOps pipelines for real-world applications.
	PSO2	1	The student is able to develop automated solutions using DevOps pipelines for real-world applications.
CO-3	PO1	2	The student is able to apply computing fundamentals to version control and containerization (Competencies:1.2.1).
	PO2	2	The student is able to identify appropriate version control strategies for software systems (Competencies:2.5.1).
	PO3	3	- The student is able to design container-based deployment solutions (Competencies:3.6.2). - The student is able to integrate modules using Docker containers (Competencies:3.8.2).
	PO4	2	The student is able to analyze system performance using containers (Competencies:4.6.1).
	PO5	3	- The student is able to demonstrate proficiency in Git and Docker tools (Competencies:5.5.2). - The student is able to evaluate strengths and limitations of containerization tools (Competencies:5.5.1).
	PO9	2	The student is able to document containerized deployment processes (Competencies:9.4.2).
	PO10	1	The student is able to manage deployment workflows

			efficiently (Competencies:10.3.2)
	PSO1	1	The student is able to implement automated and scalable deployment systems using containers.
	PSO2	1	The student is able to implement automated and scalable deployment systems using containers.
CO-4	PO1	2	The student is able to apply computing and cloud concepts for infrastructure automation (Competencies:1.2.1).
	PO2	2	The student is able to analyze infrastructure requirements and identify automation strategies (Competencies:2.5.2).
	PO3	3	- The student is able to design scalable infrastructure solutions using configuration management tools (Competencies:3.6.1). - The student is able to implement automated infrastructure workflows (Competencies:3.8.1).
	PO4	3	- The student is able to analyze infrastructure performance and optimize deployment (Competencies:4.6.1). - The student is able to interpret system results and improve configurations (Competencies:4.6.2).
	PO5	3	- The student is able to demonstrate proficiency in tools like Ansible, Puppet, Chef and SaltStack (Competencies:5.5.2). - The student is able to validate automation tools and verify results (Competencies:5.6.2).
	PO9	2	The student is able to prepare reports and documentation for automated systems (Competencies:9.4.2).
	PO10	2	The student is able to manage infrastructure deployment within constraints (Competencies:10.3.2).
	PO11	2	The student is able to update knowledge with evolving DevOps tools and technologies (Competencies:11.2.2).
	PSO1	2	The student is able to design automated systems for real-world infrastructure problems.
	PSO2	2	The student is able to design automated systems for real-world infrastructure problems.

SDG Mapping:

CO	SDG	SDG Title	Brief Linkage Explanation
CO1	SDG 4	Quality Education	Covers DevOps principles, lifecycle, and tools, enabling students to acquire practical technical skills in software development, automation, and continuous integration, supporting quality technical education and lifelong learning.
CO2	SDG 8	Decent Work and Economic Growth	Focuses on CI/CD pipelines and automation using Jenkins, which improves productivity, reduces deployment time, and enhances employability through industry-relevant DevOps skills.
CO3	SDG 9	Industry, Innovation and Infrastructure	Involves version control systems and containerization (Git, Docker), supporting scalable and reliable software infrastructure, promoting innovation and modern industrial practices.
CO4	SDG 11	Sustainable Cities and Communities	Uses infrastructure as code and configuration management tools (Ansible, Puppet, Chef) to optimize resource utilization, improve system efficiency, and enable sustainable and reliable IT infrastructure.

Credit-hour justification:

Component	Activity	Hours	Total
L	Lecture hours	15 x 1 =15	15
T	Tutorial	-	-
P	Practical hours	15x3 = 45	45
	Sub total		60
TW	Assignment 1	-	-
	Assignment 2	-	-
	Pedagogy	-	-
	Group Discussion / Seminar	-	-
	Case Study	-	-
SL	Self-Learning (video lectures / online resources MOOCs / research papers) – Virtual Labs	15x1=15	15
	Assessment / Reflection / Viva / Quiz	-	-
	Sub total		15
	Grand total		75